HIERARCHICAL MODELLING AND SIMULATION OF AN 8-BIT ARITHMETIC LOGIC UNIT USING STRUCTURAL VHDL

MINI PROJECT REPORT

ON

MODELLING AND SIMULATION OF DIGITAL SYSTEMS (ECEL-503)

Course Instructor: DR. RIKU CHUTIA

Submitted By:
HIMANSHU RABHA (ECD25012)
MELVIN BORO (ECD25013)

SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAM: ELECTRONICS DESIGN AND TECHNOLOGY

TEZPUR-784028, ASSAM, INDIA

Abstract

This mini project presents the design and simulation of an 8-bit Arithmetic Logic Unit (ALU) using Structural VHDL. The design follows a three-level hierarchical approach. At the first level, Half Adder and Full Adder modules are designed. At the second level, a 4-bit Ripple Carry Adder (RCA) is implemented using Full Adders. Finally, at the third level, an 8-bit ALU is developed by combining two 4-bit RCAs and logic circuits.

The ALU performs arithmetic and logical operations such as Addition, AND, OR, and XOR. The design is simulated using Xilinx ISE and the results verify the correctness of all operations. This project demonstrates the importance of hierarchical modeling and modular design in digital systems.

INTRODUCTION

Hardware Description Languages (HDLs) such as VHDL are widely used to model and simulate digital systems before hardware implementation. Structural VHDL is one of the design methodologies where a system is built by interconnecting smaller modules.

An Arithmetic Logic Unit (ALU) is the computational core of digital processors and embedded systems. It performs arithmetic operations and logical functions required during data processing.

In this project, an 8-bit ALU is designed using Structural VHDL and verified through simulation. The design follows a three-level hierarchy which makes the circuit modular, reusable, and easy to understand.

OBJECTIVES

The objectives of this project are:

- To understand Structural VHDL modeling.
- To learn hierarchical design methodology.
- To design an 8-bit Arithmetic Logic Unit.
- To simulate and verify digital circuits.
- To understand modular hardware development.

DESIGN METHODOLOGY

The project is implemented using three levels of hierarchy.

Level 1: Half Adder

The Half Adder is the basic building block of the design.

Inputs:

A, B

Outputs:

Sum, Carry

Equations:

Sum = A XOR B

Carry = A AND B

Level 2: Full Adder

A Full Adder is constructed using two Half Adders and one OR gate.

Inputs:

A, B, Cin

Outputs:

Sum, Cout

The Full Adder is capable of adding three binary inputs and producing a sum and carry output.

Level 3: 4-Bit Ripple Carry Adder

Four Full Adders are connected in series to create a 4-bit Ripple Carry Adder.

Features:

- Binary addition of two 4-bit numbers.
- Carry propagates through each stage.
- Simple and modular architecture.

Top Level: 8-Bit ALU

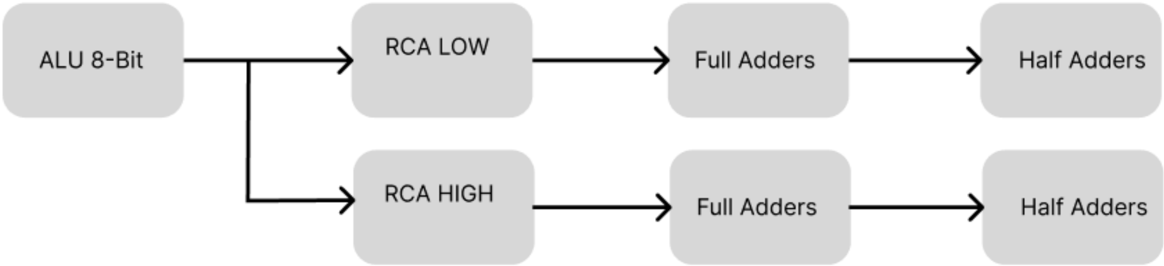
The final ALU is built using two 4-bit Ripple Carry Adders.

Supported operations:

SEL	OPERATION
00	Addition
01	AND
10	OR
11	XOR

HIERARCHICAL ARCHITECTURE

The complete hierarchy of the design is shown below:



This hierarchy clearly demonstrates modular design and component reuse.

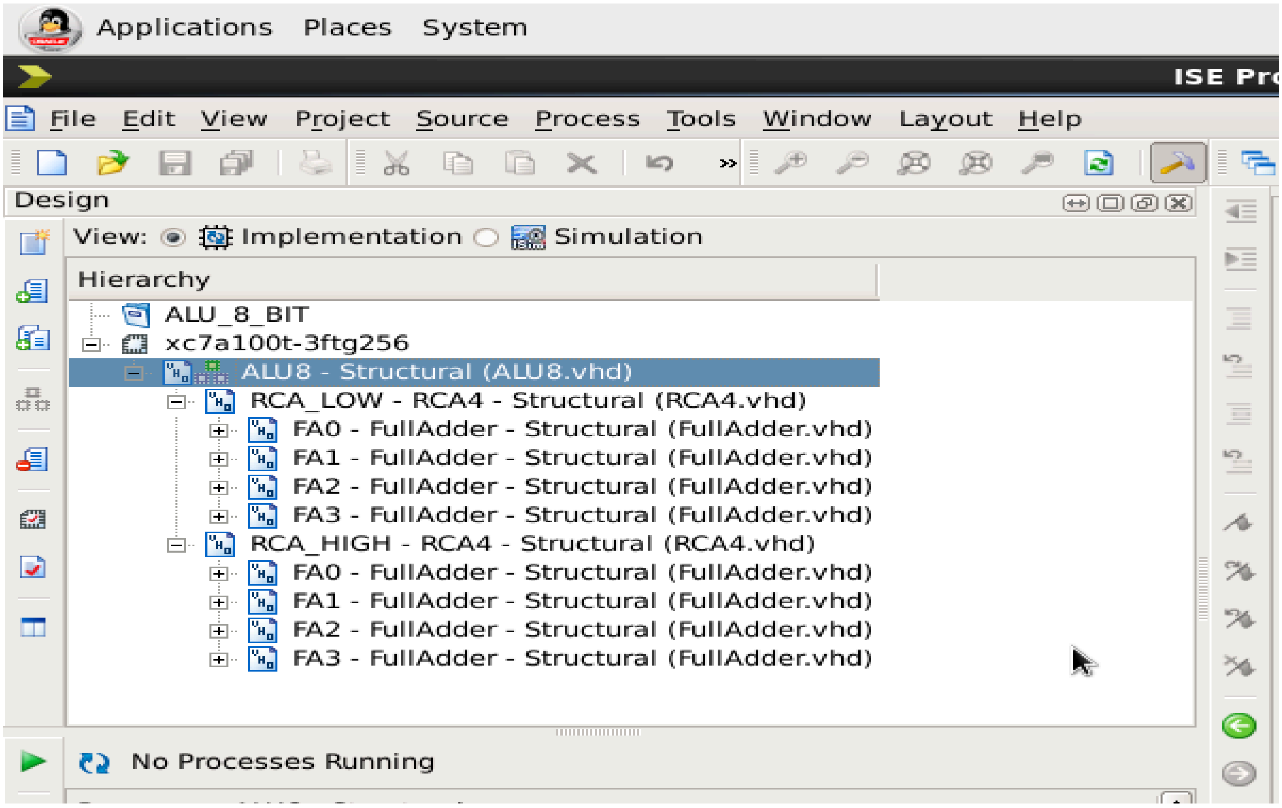


Figure 1: Hierarchy Viewer Generated in Xilinx ISE .

VHDL IMPLEMENTATION

The project consists of five VHDL files:

1. HalfAdder.vhd
2. FullAdder.vhd
3. RCA4.vhd
4. ALU8.vhd
5. tb_ALU8.vhd

The Half Adder forms the basic module. Two Half Adders are used to construct a Full Adder. Four Full Adders are combined to create a 4-bit Ripple Carry Adder. Two Ripple Carry Adders are then used to implement the 8-bit arithmetic section of the ALU.

The logical operations (AND, OR, XOR) are implemented using VHDL operators. A selection signal determines which operation is performed.

SIMULATION SETUP

Simulation was performed using Xilinx ISE Design Suite.

Steps followed:

1. Create VHDL source files.
2. Compile all modules.
3. Create a testbench.
4. Apply test inputs.
5. Run behavioral simulation.
6. Verify outputs using a waveform viewer.

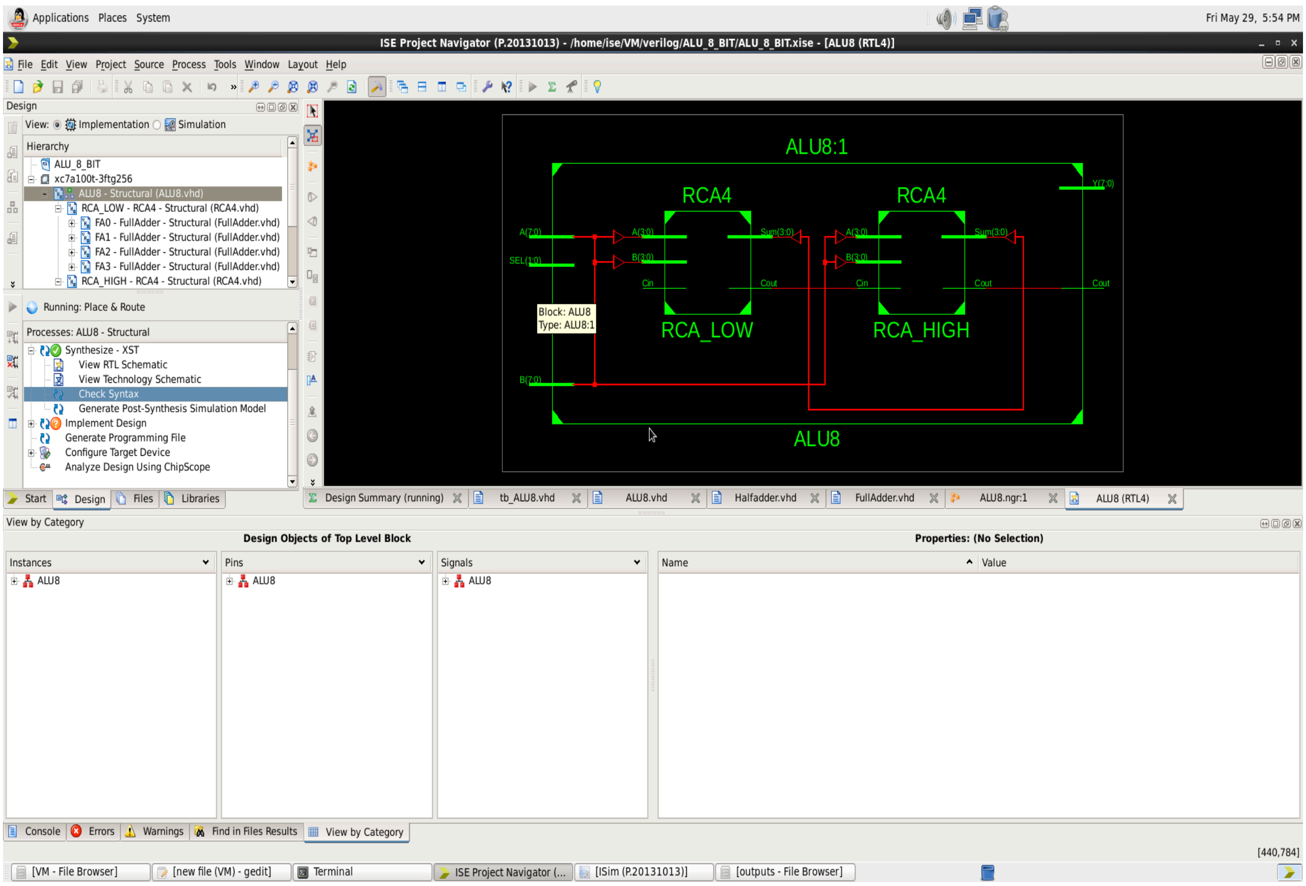


Figure 2: RTL Schematic of 8-Bit ALU .

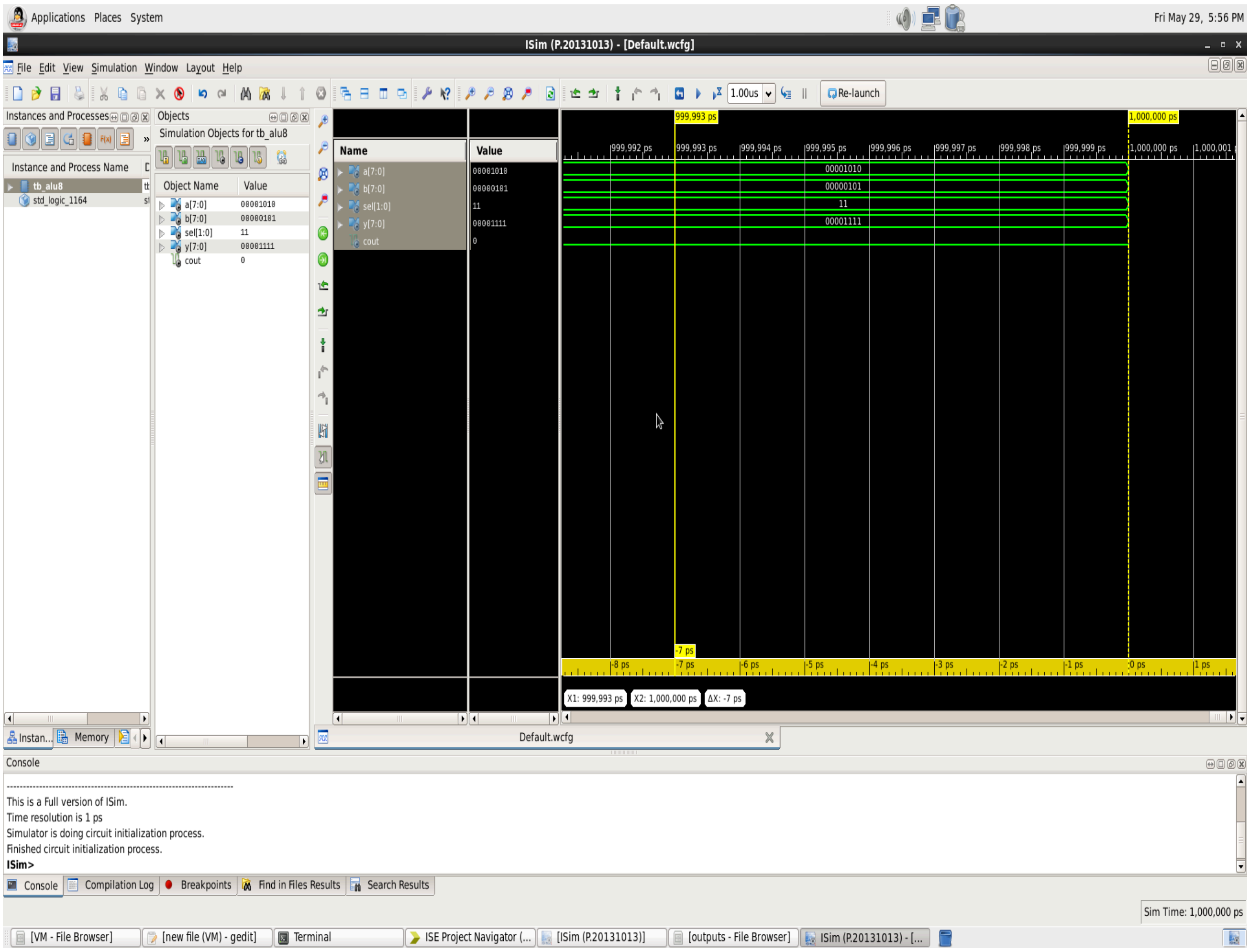


Figure 3:Functional Simulation Results .

RESULTS AND DISCUSSION

The following test inputs were applied:

A = 10

B = 5

The simulation results are shown below.

SEL	**OPERATION**	**OUTPUT**
00	*ADDITION*	15
01	*AND*	0
10	*OR*	15
11	*XOR*	15

The simulation outputs matched the expected theoretical values. The hierarchy viewer also confirmed the proper structural implementation of the design.

ADVANTAGES

- Modular design.
- Easy debugging.
- Reusable components.
- Simplified verification.
- Scalable architecture.
- Better understanding of digital design concepts.

APPLICATIONS

The Arithmetic Logic Unit is widely used in:

- Microprocessors
- Microcontrollers
- Embedded Systems
- FPGA-Based Systems
- Digital Signal Processing Applications
- Arithmetic Processing Units

CONCLUSION

An 8-bit Arithmetic Logic Unit was successfully designed and simulated using Structural VHDL. The design follows a three-level hierarchical architecture consisting of Half Adders, Full Adders, Ripple Carry Adders, and the final ALU module.

Simulation results confirmed the correct operation of arithmetic and logical functions. The project helped in understanding hierarchical modeling, structural design techniques, and digital system simulation using VHDL.